

~~X~~ Variables

In this JS, variables are used for storing data values. It is like a container in JS which holds different types of info.

To declare a variable in JS we use "var", "let" or "const" keyword, followed by variable name.

(1) "Var" - The "var" keyword is the main way for declaring variables in JS. It is accessible within the entire function where it is declared.

Example -

```
// declaring & assigning :- variable using var  
1. var age = 20;  
2. console.log(age);
```

• A variable can be assigned a data value using the assignment op. equal sign (`=`).

```
// Reassigning the value of variable
```


3. `a = 22;`

4. `console.log (age);`

- A variable can be reassigned in JS. by assigning a new value to the variable without using the "var" keyword.

(ii) "let" → the "let" keyword was introduced to address the issues within "var" keyword & is accessible within the nearest block where it is declared.

Example -

```
// declaring & assigning a variable
// using let
```

```
let name = "Saurabh Anand";
console.log (name);
```

```
{ // output : Saurabh Anand }
```

```
// Block scope with let
```

```
if (true) {
    let num = 20;
    console.log (num);
}
```

```
{ // output : 20 }
```


(iii) "const" → The "const" stands for "constant". It has also block scope like "let", but in "const" declaration of variable cannot be reassigned after it has been assigned a value.

Example -

// declaring a "const" variables.

```
const PI = 3.14; { PI = 3.14 }  
console.log("PI" + PI);
```

XX Data Types

Data types are used for holding & manipulating the values in the JS. The common data types are numbers, strings, boolean, undefined, null, array, object, function & symbols. Each data type has specific behaviour & properties.

(P) Number → It is used for representing the numeric value in the JS.

Example -

```
let age = 10;
```

```
let temperature = 99 - 1;
```

→ we can perform arithmetic op. also on no. values

- (ii) String - It is used for representing the sequence of character values in J.S. It is enclosed within single quotes or double quotes.

Example -

let name = "Saurabh Anand";

- (iii) Boolean → It is used for representing the logical value that can either be true or false.

Example -

let has-Permission = true;

- (iv) Undefined → It is used for representing value or variable that has been declared but has not been assigned a value.

Example -

let x;

- (v) NULL → It is used for representing the intentional absence of any object value.

Example -

let data = null;

(vi) Object → It is complex data type which represents the collection of properties and their value. It is used for organizing and store related information for functioning together.

Example -

```
let person = {  
  id : "Abhishek",  
  name : 20,  
  is-Student : true  
};
```

(vii) Array → Used for representing an order list of values. It can store multiple values of any data type, such as strings, objects, numbers.

Example -

```
let arr = ['saureabh', [], {4, () => {3, 1]}
```

(viii) Function → treated as first class objects, we can store functions in variables, pass arguments to other functions & can return functions.

so technically, the data type of a fun. is an object.